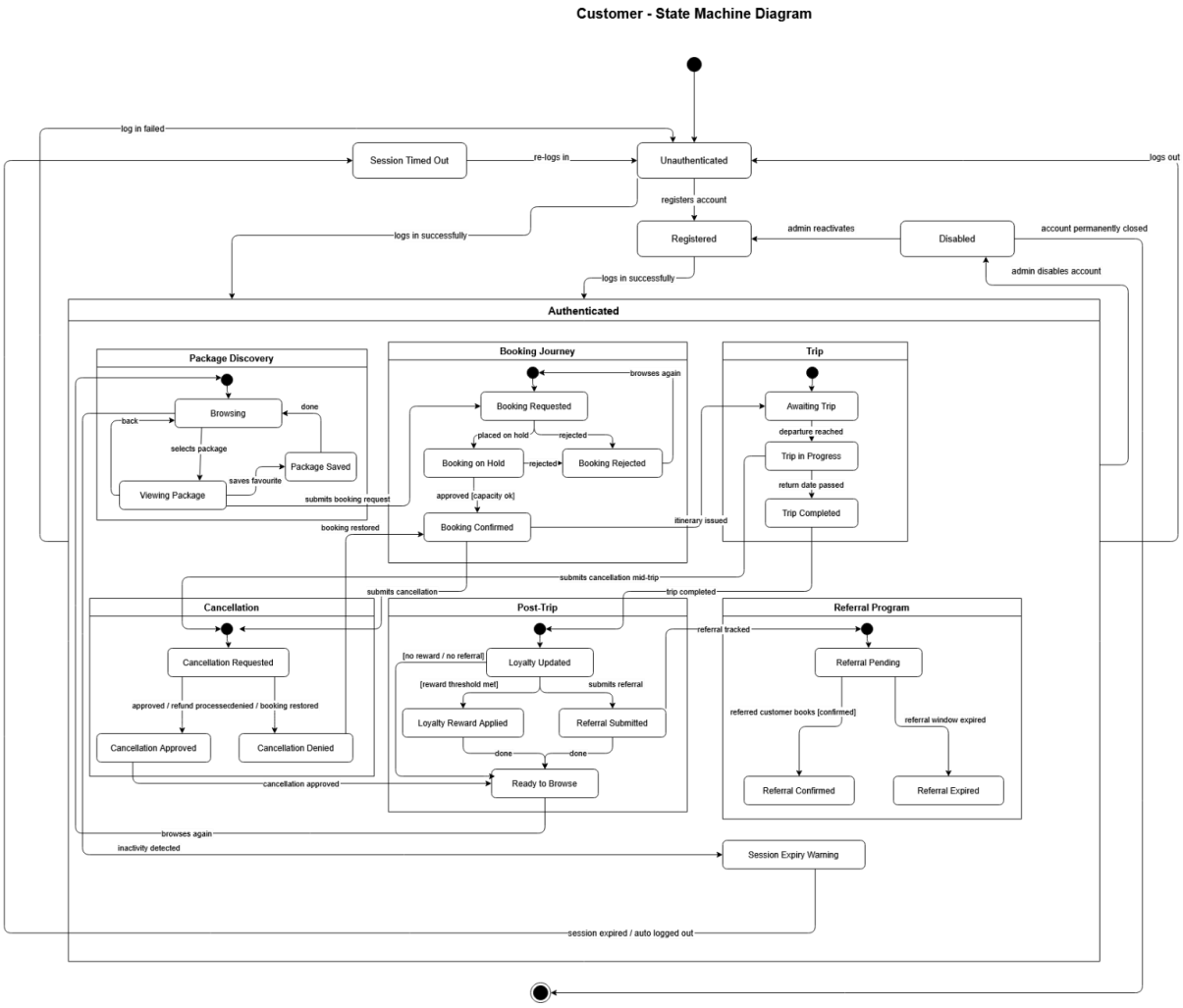
State Machine Diagrams

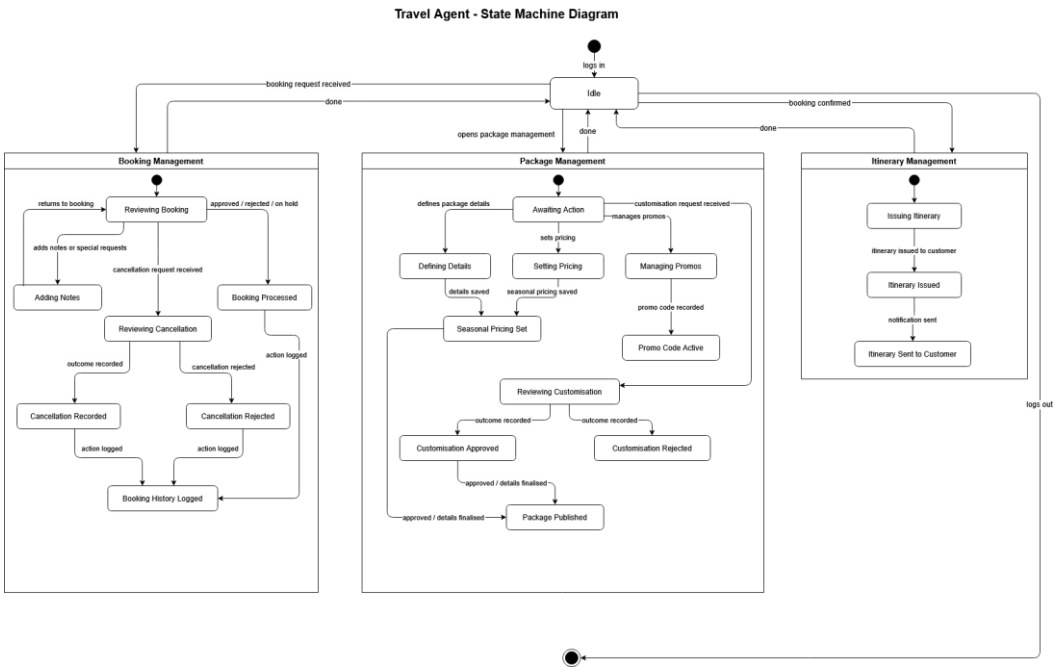
1. Customer

The Customer moves through five composite regions within an Authenticated state: Package Discovery, Booking Journey, Trip, Cancellation, and Post-Trip. Package Discovery covers browsing, viewing, and saving packages. Booking Journey handles the full request-to-confirmation flow including hold and rejection paths. The Trip region tracks departure through completion. Cancellation handles mid-booking and mid-trip requests with approval and denial outcomes. Post-Trip models loyalty reward application and referral submission, including a dedicated Referral Program region with pending, confirmed, and expired states. Session timeout and account disable/reactivate paths are modeled outside the Authenticated boundary.



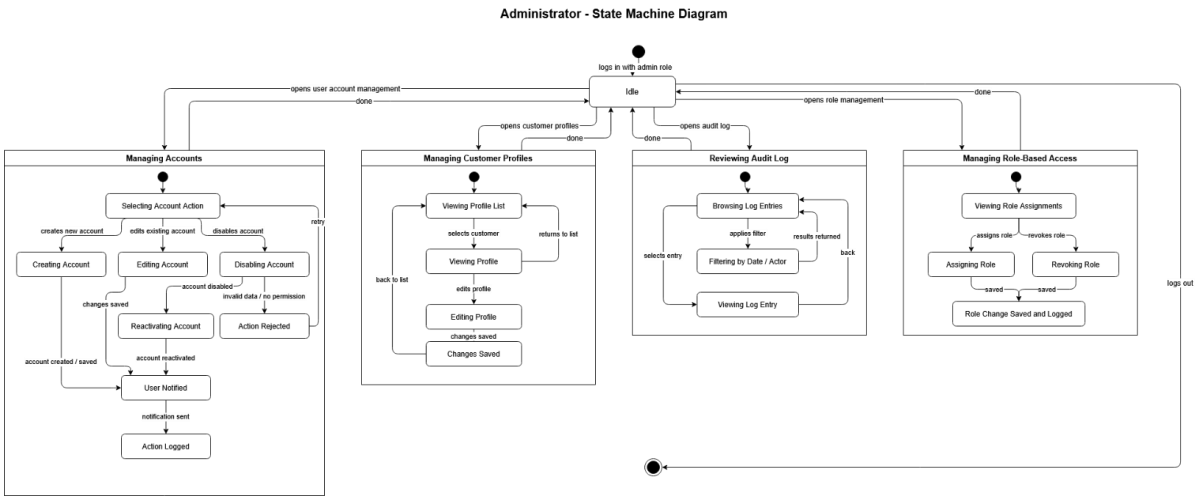
2. Travel Agent

The Travel Agent works across three composite regions accessible from a central Idle state: Booking Management, Package Management, and Itinerary Management. Booking Management covers reviewing bookings, adding notes, processing decisions, handling cancellation requests, and logging booking history. Package Management covers defining package details, setting seasonal pricing, recording promo codes, and reviewing customisation requests through to publication. Itinerary Management covers issuing and delivering the digital itinerary to the confirmed customer.



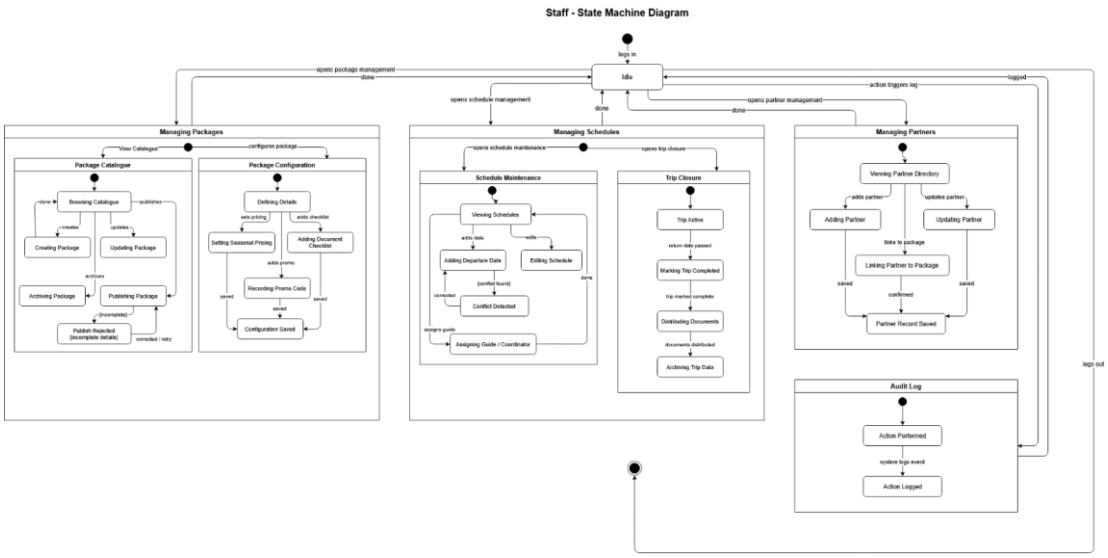
3. Administrator

The Administrator operates across four regions from a central Idle state: Managing Accounts, Managing Customer Profiles, Reviewing Audit Log, and Managing Role-Based Access. Account management covers the full lifecycle creating, editing, disabling, and reactivating accounts with validation error handling and user notification. Customer profile management allows viewing and editing individual profiles. The audit log region supports browsing, filtering, and inspecting individual log entries. Role management covers assigning and revoking access roles, with all changes saved and logged.



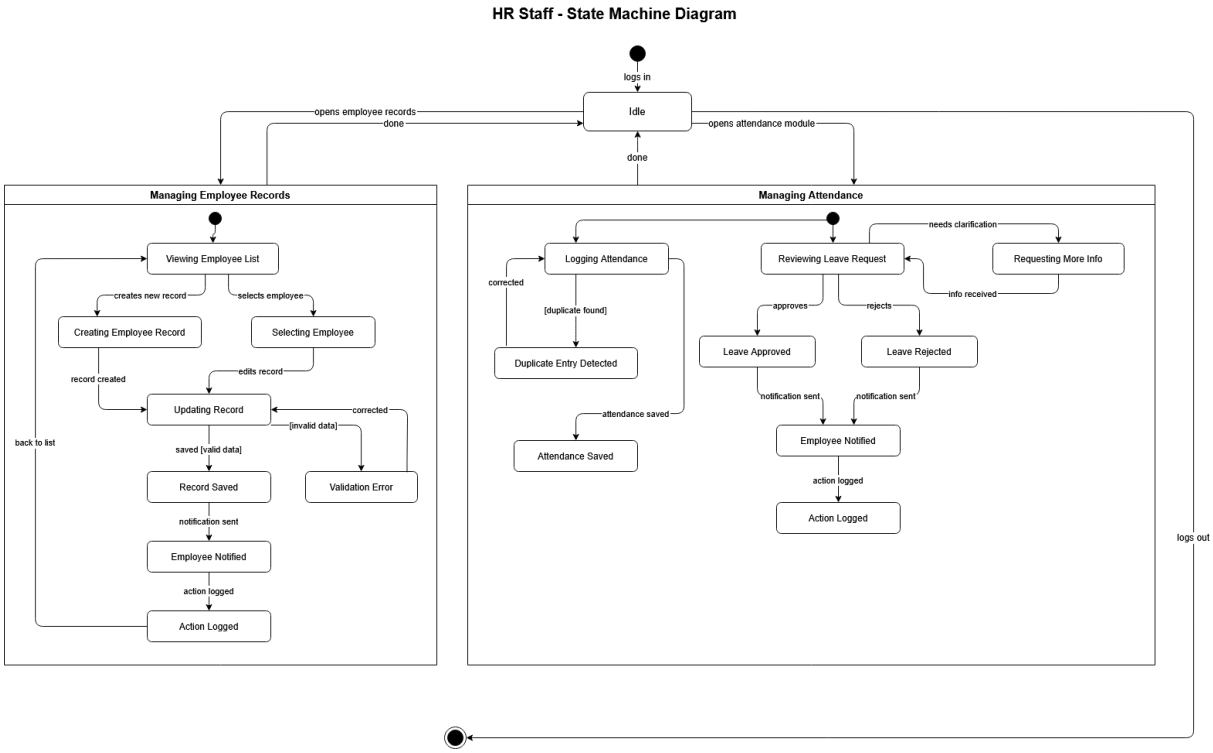
4. Staff

Staff is the broadest operational actor in the system, working across four regions: Managing Packages, Managing Schedules, Managing Partners, and Audit Log. Managing Packages is subdivided into Package Catalogue, covering create, update, archive, and publish actions, and Package Configuration, covering details, seasonal pricing, document checklists, and promo codes. Managing Schedules is subdivided into Schedule Maintenance, covering departure dates, conflict detection, and guide assignment, and Trip Closure, covering completion marking, document distribution, and archiving. Partner management handles the business partner directory and package linking. All actions trigger an audit log entry.



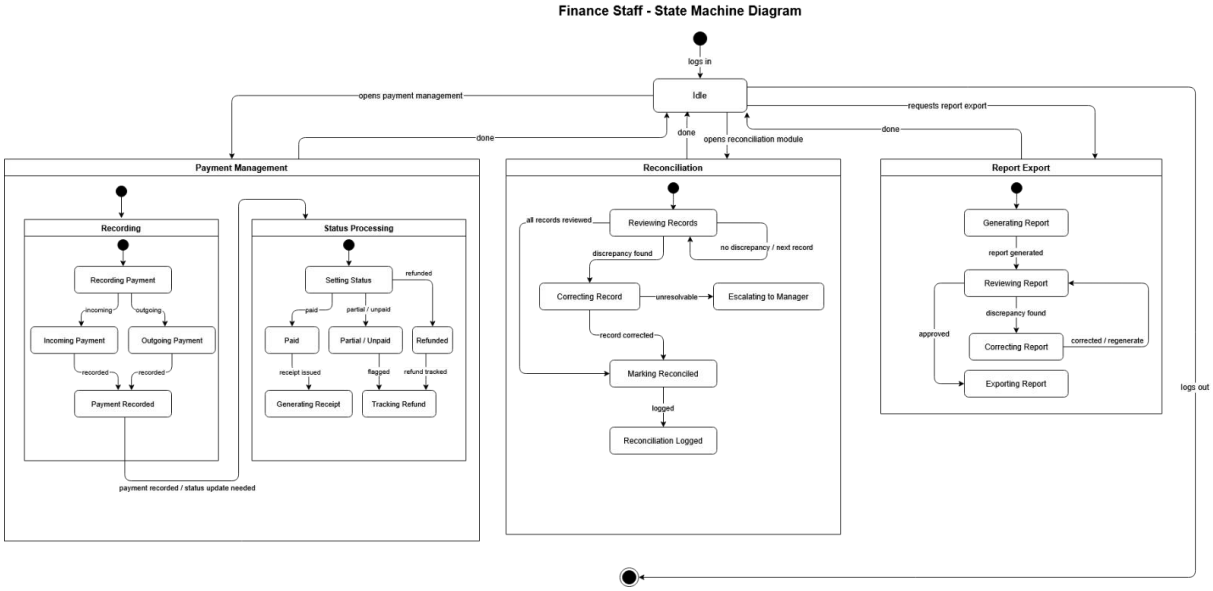
5. HR Staff

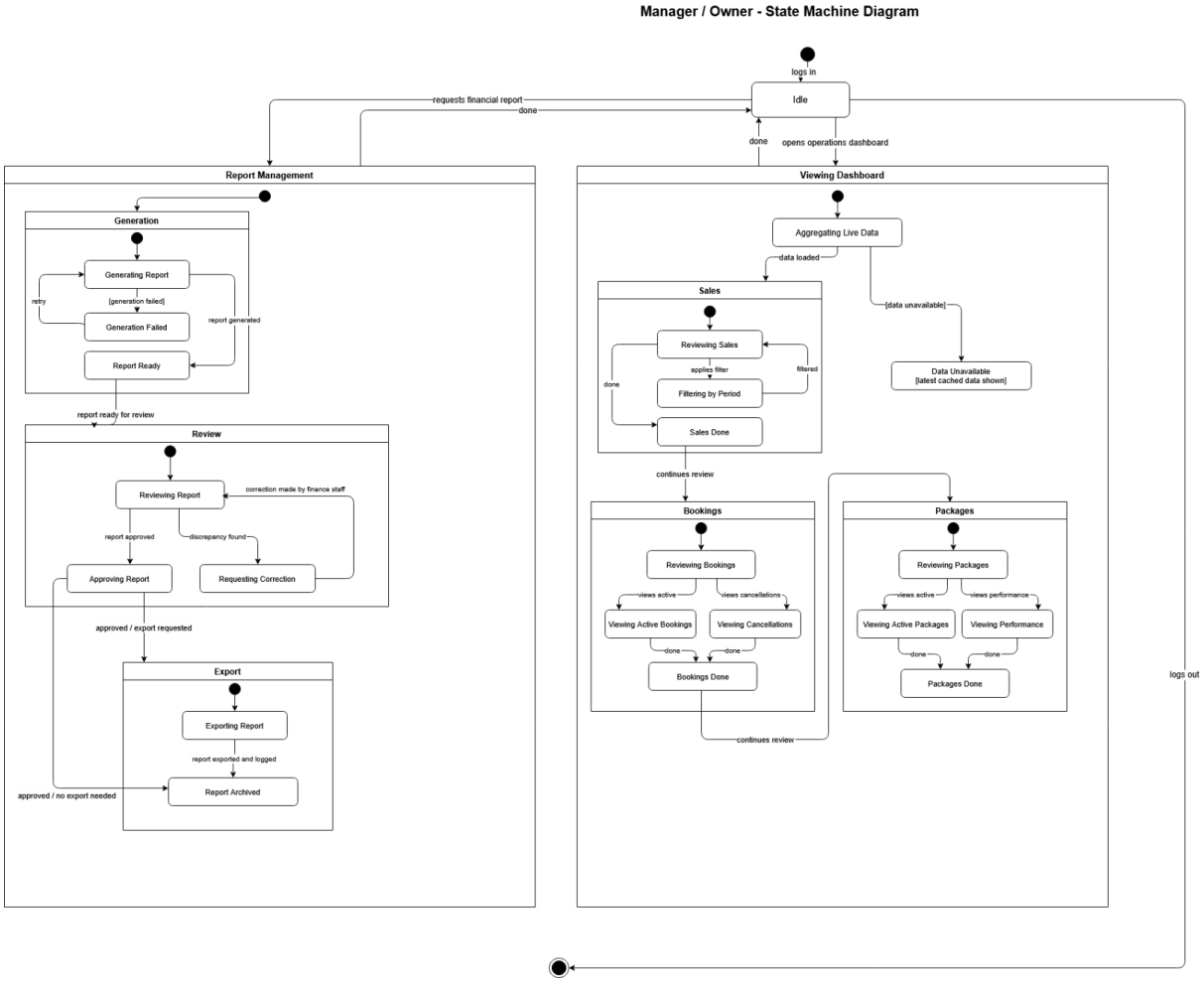
HR Staff work across two regions from a central Idle state: Managing Employee Records and Managing Attendance. Employee records management covers creating new records, selecting and updating existing ones, with validation error handling, notification, and action logging. Attendance management covers logging attendance entries with duplicate detection, and a leave request workflow that includes a clarification step before approve or reject decisions, followed by employee notification and action logging.



6. Finance Staff

Finance Staff work across three regions from a central Idle state: Payment Management, Reconciliation, and Report Export. Payment Management is subdivided into Recording, which distinguishes incoming from outgoing payments, and Status Processing, which branches across paid, partial/unpaid, and refunded outcomes triggering receipt generation, refund tracking, or follow-up flagging. Reconciliation covers record review, discrepancy correction, and escalation to management for unresolvable issues, followed by reconciliation logging. Report Export covers generation, review, correction, and download, with the export action logged to the audit trail.



7. Manager / Owner

The Manager works across two regions from a central Idle state: Report Management and Viewing Dashboard. Report Management is subdivided into Generation, which handles report creation with a retry path on failure, Review, which covers approval and correction requests, and Export, which handles download and archiving with an option to skip export when not needed. Viewing Dashboard is subdivided into Sales, Bookings, and Packages, each with their own internal review flow. Sales includes period filtering. Bookings separates active bookings from cancellations. Packages separates active package review from performance review.